

1. Palindrome

✓
r/string

✓
bers

✓
quivalent to
l octal

✓
nd nth min
en sum of it

✓
positive and
upto -1

✓
es in array

Q1 Palindrome number/string

10 marks

Palindrome number/string

Your Code

```
1 import java.util.*;
2 import java.lang.*;
3 public class main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         String s=sc.nextLine();
7         String rev="";
8         int i;
9         for(i=s.length()-1;i>=0;i--){
10             rev+=s.charAt(i);
11         }
12         if(rev.equals(s)){
13             System.out.println("its a palindrome");
14         }
15         else{
16             System.out.println("not a palindrome");
17         }
18     }
19 }
```

Run Save

Input

12344321

Output

its a palindrome

2. First n perfect numbers

✓
Q1
Palindrome number/string
10 marks

✓
Q2
First n perfect numbers
10 marks

✓
Q3
Decimal number equivalent to
binary number and octal
number
10 marks

✓
Q4
Nth max number and nth min
number in array then sum of it
and difference of it
10 marks

✓
Q5
Find the average positive and
negative numbers upto -1
10 marks

✓
Q6
Removing duplicates in array
10 marks

✓
Q7

Q2 First n perfect numbers

10 marks

First n perfect numbers

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int num=sc.nextInt();
6         for(int j=1;j<num;j++){
7             int sum=0;
8             for(int i=1;i<j;i++){
9                 if(j%i==0){
10                     sum+=i;
11                 }
12             }
13             if(sum==j){
14                 System.out.print(j+" ");
15             }
16         }
17         System.out.println();
18         sc.close();
19     }
20 }
```

Run Save

Input

500

Output

6 28 496

3. Removing duplicate elements

Removing duplicates in array

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         String arr={"mango","papaya","gooseberry","melon","mango","me
5         String temp[]=new String[arr.length];
6         int count=0;
7         int i,j;
8         for(i=0;i<arr.length;i++){
9             boolean isduplicate=false;
10            for(j=0;j<count;j++){
11                if(arr[i].equals(temp[j])){
12                    isduplicate=true;
13                    break;
14                }
15            }
16            if(isduplicate==false){
17                temp[count]=arr[i];
18                count++;
19            }
20        }
21        for(i=0;i<count;i++){
22            System.out.print(temp[i]+" ");
23        }
24    }
```

Input

stdin input

Output

mango papaya gooseberry melon apple
kiwi

4. Max and Min number and the sum of it and difference

Q4 Nth max number and nth min number in array then sum of it and difference of it

10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=sc.nextInt();
9         }
10        int lar=sc.nextInt();
11        int small=sc.nextInt();
12        Arrays.sort(arr);
13        int largest=arr[n-lar];
14        int smallest=arr[small-1];
15        int op=largest-smallest;
16        int sp=largest+smallest;
17        System.out.println("add is: "+sp);
18        System.out.println("sub is: "+op);
19    }
20 }
```

Input

5
3 5 6 8 9
2
4

Output

add is: 16
sub is: 0

5. Positive and Negative average until -1 is pressed

Your Code

```

1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int i,pos=0,neg=0,p=0,ni=0;
6         while(true){
7
8             int num=sc.nextInt();
9             if(num==1){
10                 break;
11             }
12             else if(num>0){
13                 p+=num;
14                 pos++;
15             }
16             else{
17                 ni+=num;
18                 neg++;
19             }
20         }
21         double pavg=(double)p/pos;
22         double navg=(double)ni/neg;
23         System.out.println("positive average: "+pavg);
24         System.out.println("negative average: "+navg);
25
26     }
27 }

```

Input

6
1
2
3

Output

```

positive average: 3.0
negative average: -3.0

```

6. Decimal to Binary and octal

Your Code

```

1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int deci=sc.nextInt();
6         int d=deci;
7         int deez=deci;
8         int bin=0,place=1;
9         int r;
10        while(d>0){
11            r=d%2;
12            bin=bin+r*place;
13            place=place*10;
14            d=d/2;
15        }
16        System.out.println("binary is: "+bin);
17        int oct=0,base=1;
18        while(deez>0){
19            int temp=deez%8;
20            oct=oct+temp*base;
21            base=base*10;
22            deez=deez/8;
23        }
24
25        System.out.println("octal is: "+oct);
26    }
27 }
28 }

```

Input

72

Output

```

binary is: 1001000
octal is: 110

```

7. Matrix Multiplication

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int i,j,k;
6         int n=sc.nextInt();
7         int a[][]=new int[n][n];
8         int b[][]=new int[n][n];
9         int c[][]=new int[n][n];
10        for(i=0;i<n;i++){
11            for(j=0;j<n;j++){
12                a[i][j]=sc.nextInt();
13            }
14        }
15        for(i=0;i<n;i++){
16            for(j=0;j<n;j++){
17                b[i][j]=sc.nextInt();
18            }
19        }
20        for(i=0;i<n;i++){
21            for(j=0;j<n;j++){
22                c[i][j]=0;
23                for(k=0;k<n;k++){
24                    c[i][j]+=a[i][k]*b[k][j];
25                }
26            }
27        }
28        for(i=0;i<n;i++){
29            for(j=0;j<n;j++){
30                System.out.print(c[i][j]+" ");
31            }
32            System.out.println(" ");
33        }
34        sc.close();
35    }
36 }
```

Input

```
2
1 2
3 4
5 6
```

Output

```
13 12
31 30
```